

OPERATING MANUAL

MHS-5200A

Dual-channel

DDS Signal Generator

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Unpacking

When you get a new MHS-5200A Series dual-channel DDS signal generator, it is recommended that you follow these steps to inspect the instrument.

1. Check for transportation damage caused.

Such as packing or bubble bag cushions serious damage, keep them until the machine and accessories passed the test.

2. Check the box in the article are complete.

The package contents as described below. If the content does not match or if the instrument is damaged, contact your dealer or manufacturer.

Host computer:	MHS-5200A Series dual-channel DDS signal generator	1
Accessory:	Power Adapter	1
	USB cable	1
	Signal connection cable	2
	TTL-Ext. connector	1
	User Manual (pdf version) and Software on CD	1

3. Check the machine

Inspect the instrument is damaged, not working properly, or fails performance tests, please contact your dealer or the Company.

1. Outline

1-1. The instrument Introduction

MHS-5200A series instruments using large scale integrated circuits and high-speed FPGA MCU microprocessor, the internal circuit to take surface mount technology has greatly enhanced the instrument's noise immunity and service life. Display interface using LCD1602 display is divided into two lines, the top line shows the current frequency, the following line displays additional parameters or function variable and flexible use of flip key setting, greatly enhances the operability. This instrument signal generation, waveform scanning, as well as the use of parameter measurement has great advantages, is an electronic engineer, electronic laboratories, production lines and teaching, research and the ideal test, measurement equipment.

Model Description

This series of instruments in four models, the main difference is the maximum frequency sine wave, as follows:

MHS-5200-06M sinusoidal signals at frequencies up to 6MHz

MHS-5200-12M sinusoidal signals at frequencies up to 12MHz

MHS-5200-20M sinusoidal signals at frequencies up to 20MHz

MHS-5200-25M sinusoidal signals at frequencies up to 25MHz

The instrument characteristics

1. Direct digital synthesis (DDS) technology, FPGA design, low power consumption;
2. Dual output, can work in sync phase adjustable;
3. It has up to 999 seconds of linear and logarithmic sweep function Sweep;
4. With a sine wave, triangle wave, square wave, sawtooth rise, falling sawtooth waveform basic function and variable duty cycle pulse wave, but also has 16 sets of arbitrary waveform customized by the user;
5. A total of 10 sets of parameters stored bits have M0 ~ M9, M0 boot automatically transferred out of data;
6. At 12MHz or less, the sharpest of up to 20Vp-p, more than 12MHz, the biggest reach 15Vp-p;

7. Built precision -20dB attenuator reach the minimum amplitude resolution 1mV;
8. With a 120% ~ + 120% DC bias function;
9. Pulse Duty precise adjustment to 0.1%;
10. Having four variable phase difference of TTL output;
11. Having a frequency measurement, period measurement, positive and negative pulse width measurement, duty cycle measurements and counting function;
12. Four optional gate frequency measurement time, which strike a balance between speed and accuracy;
13. All parametric equalizer can be done by an internal calibration procedure;
14. Powerful communications features, completely open communication protocol that allows the secondary development of very simple;
15. After connecting with the PC, the PC can be used to control the instrument, and can be edited on the PC arbitrary waveform output waveform is downloaded to the instrument;
16. This type machine can be equipped with an increase in power module, the signal output amplitude reaches 30Vpp, the maximum output current reaches 1A.

Technical indicators

Table 1-1 MHS-5200A Series Specifications

Project		Parameters	
Key Features	Frequency Range	Sine wave	Normal mode: MHS-5200-06M: 0Hz ~ 6MHz; MHS-5200-12M: 0Hz ~12MHz; MHS-5200-20M: 0Hz ~ 20MHz; MHS-5200-25M: 0Hz ~ 25MHz-
		Square wave	0Hz ~ 6MHz
		Triangle wave	0Hz ~ 6MHz
		Sawtooth	0Hz ~ 6MHz
		Arbitrary Waveform	0Hz ~ 6MHz
		TTLDigital signal wave	0Hz ~ 6MHz
	Output modulation	Frequency sweep	
	Waveform types	Sine, square, triangle, sawtooth lift, TTL digital signal wave, Arb	
	Waveform Length	2048 Point	

	Sampling rate	200MSa/s
	Waveform amplitude resolution	12 bits
	The minimum frequency resolution	10mHz
	Frequency error	$\pm 5 \times 10^{-6}$
	Frequency stability	$\pm 1 \times 10^{-6}$
	Amplitude range (peak to peak)	5mVp-p ~ 20Vp-p (12MHz or less) 5mVp-p ~ 15Vp-p (12MHz above)
	Output Impedance	50Q $\pm$ 10%
	Amplitude resolution	1mVp-p (-20dB Attenuation) 10mVp-p (Does not decay)
	Amplitude stability	$\pm 0.5\%$ (Every five hours)
	Amplitude error	$\pm 1\% + 10\text{mV}$ (Frequency 1KHz, 15Vp-p)
	Offset Range	-120% ~ +120% (The ratio of the bias voltage and signal amplitude)
	Bias Resolution	1%
	Phase range	0 ~ 359°
	Phase resolution	1°
Sine wave	Harmonic arrived System	40dBc (<1MHz), 35dBc (1MHz ~ 20MHz)
	Distortion	<0.8% (20Hz ~ 20KHz)
Square wave	Lifting along time	<20ns
	Overshoot	<10%
	Duty cycle adjustment range	0% ~ 99.9%
TTL	Lifting along time	<20ns
	LOW	<0.3V
	High	1V ~ 10V
Arbitrary Waveform	Quantity	16 groups
	Memory depth / group	1KB /16 group
Scan	Scan Mode	Linear sweep, log sweep
	Scan time	1S ~ 999S
	Scan range	It is determined by the sweep parameter settings
External measurements	Frequency range	GATE-TIME=10S 0.1HZ ~ 60MHZ GATE-TIME= 1S 1HZ ~ 60MHZ GATE-TIME=0.1S 10HZ ~ 60MHZ GATE-TIME=0.01S 100HZ ~ 60MHZ

	Input voltage range	0.5Vp-p ~ 20Vp-p
	Counting range	0 ~ 429 496 7295
	Counting	Manually
	Positive and negative pulse width measurement	10ns resolution, the maximum measurable 10s
	Periodic measurements	20ns resolution, the maximum measurable 20s
	Duty Cycle Measurement	0.1% resolution, measuring range from 0.1% to 99.9%
	Source Selection	1.Ext.IN input (AC signal), 2.TTL_IN input (digital signal)
Memory	Memory	10
	Location	M0 - M9
Interface	Interface	Using USB to serial interfaces
	Communication rate	57600bps
	Protocol	Using the command line, the agreement public
Power supply	DC	DC 5V
Size	Length x width x height	180x190x71mm
Weight	Single	546g

2. Instrument Description

1. Panel

MHS-5200A appearance **Figure 2-1** shows the description of the parts as shown in **Table 2-1**.

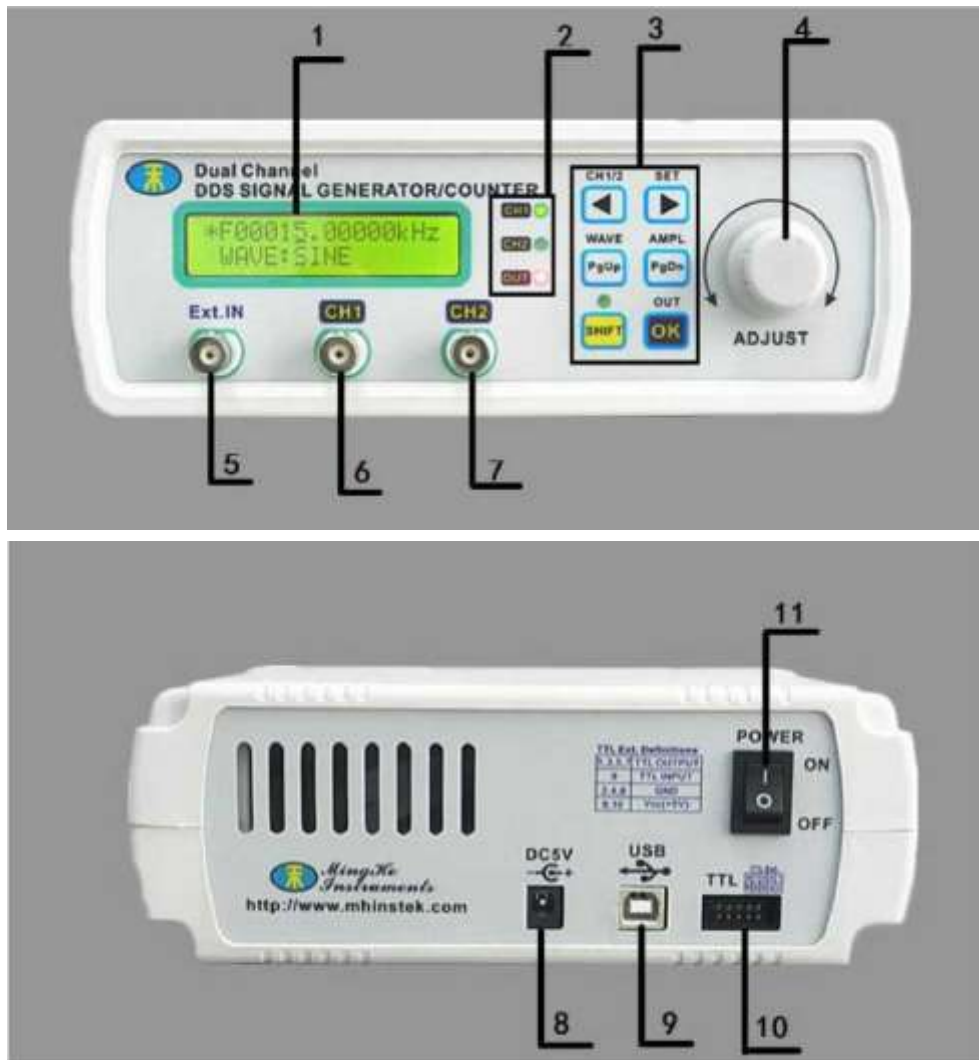


Figure 2-1 MHS5200A panel

Table 2-1 MHS5200A Panel

Grade	Explanation	Grade	Explanation
1	LCD1602	7	CH2 output interface
2	Status Indicator	8	DC5V power input
3	Operation buttons	9	USB
4	Knob	10	TTL-Ext. input / output interface
5	Ext.In input interface		
6	CH1 output interface	11	Power Switch

2. Ribbon Description

LCD instrument display is divided into two functional areas, as shown in **Figure 2-2**, description of the parts shown in **Table 2-2**.

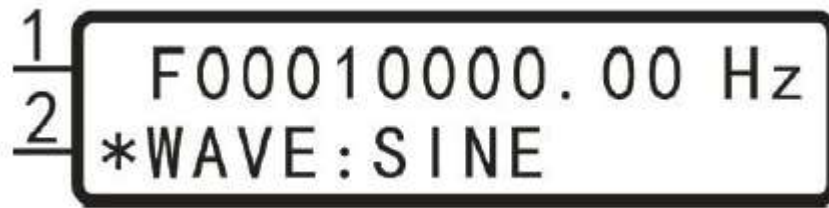


Figure 2-2. MHS5200A schematic display

Table 2-2 MHS5200A Ribbon Description

Grade	Ribbon Description
1	Frequency Display
2	Operation Feature Tip

3. Key Functions

	Cursor left adjustable parameters of the step value
	Move the cursor right step to reduce the adjusted parameter value
	Page Up key function selection
	Function selection Page Down
	OK button
	Channel switch (CH1 \ CH2)
	Switching position, when the “*” second line adjustment options to adjust the frequency of the first line
	Click this button to quickly enter a combination of waveform adjustment
	Click this button to quickly enter a combination of amplitude modulation
	Click this button to turn on or off output combination

4. Menu Function Description

1	F00015.00000KHz	It indicates the current frequency of the output waveform
2	WAVE:SINE	WAVE mean waveform, SINE represents a sine wave
3	WAVE:SQUARE	SQUARE represents a square wave
4	WAVE: TRIAN GLE	TRIANGLE triangular wave representation
5	WAVE: SAWTOOTH-R	SAWTOOTH-R represents liter sawtooth
6	WAVE: SAWTOOTH-F	SAWTOOTH-F represents drop sawtoot
7	WAVE:ARB0	ARB represents any wave, any wave 0 0 indicates the location of storage, a total of 0-15 Group on Arbitrary wave
8	AMPL: 05.00V	AMPL represents the output waveform of the peak to peak (voltage)
9	OFFS: 000%	OFFS is an offset function, -120% to + 120% adjustment
10	DUTY: 50.0%	DUTY is to adjust the duty cycle function
11	PHASE: 000°	PHASE represents the phase difference between channel 1 and channel 2
12	TRACE: OFF	Channel 1 and channel 2 track is open, OFF means closed, ON means open, the channel will change with the change of channel 1 and 2 open.
13	FREQ-UNIT :KHz	Indicates the output frequency of the unit, then the unit is KHz, press the OK button to switch.

14	INVERT: OFF	A reverse function key, you can make the output waveform phase reverse.
15	BURST: OFF	It represents burst features on or off
16	MSR-SEL:Ext.IN	Represents the input port select frequency measurement signal, Ext.IN indicates an analog signal input port, TTL.IN represents digital signal input port
17	MSR-MODE:FREQ	Represents a measurement mode, FREQ represented in this case the measurement frequency, COUNTR represents counting function, POS-PW positive pulse width measurement, NEG-PW negative pulse width measurement, PERIOD measurement period DUTY duty cycle measurements
18	GATE—TIME: 1S	Set the gate time, you can press the OK button to switch
19	F=0Hz	It represents the frequency of the waveform being measured
20	SET SWEEP FRWQ1	Indicates the sweep start frequency, set on a single line
21	SET SWEEP FREQ2	Indicates the sweep stop frequency, set on a single line
22	SWEEP TIME:001S	It indicates the sweep time
23	SWEEP MODE:LINE	Sweep mode, LINE represents a linear sweep, LOG logarithmic sweep
24	SWEEP:OFF	Sweep switch, OFF means closed, ON means to open
25	SAVE:M0	Save parameters, select the encoder switching group 10 storage locations
26	LOAD:M0	Load parameters, select the encoder switching group 10 storage locations

3. Instructions

Power

1. Access 5V supply. You can use the box to configure oriented instrument powered DC5V power adapter.
2. Enter the main interface.

Instructions

This section will detail how to operate the instrument. It should be noted that, similar to the instrument channel CH1 CH2 channel with which the operating instructions section 1-6 also apply to CH2 channel.

1. Set the CH1 waveform



Figure 2-3

In the main interface shown in **Figure 2-3**, when the "*" in the first line,

click  Bond Adjust the output waveform type, the output waveform types are sine, square, triangle, rising sawtooth, descending sawtooth wave and a 16-Group on

Arbitrary, press  Key it is possible to return ahead of the original set of waveform.

If you want to quickly adjust the output waveform can click  + , then a "*" to switch to the second row, rotating "ADJUST" Knob, you can quickly switch the output waveform types.

2. Set the CH1 frequency

In the main interface, shown in **Figure 2-4** when the "*" in the first line, by

adjusting  or  move the cursor to adjust the frequency step size, then to adjust the frequency of the output Waveform by rotating the "ADJUST" Knob.

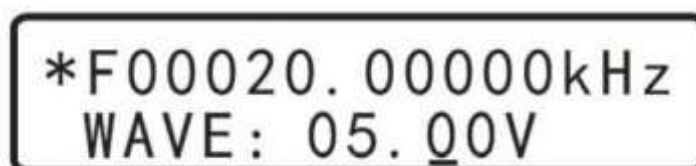


*F00020.00000kHz
WAVE: SQUARE

Figure 2-4

3. Setting the amplitude of CH1

In the main interface, press  +  Button after, the magnitude of the interface will appear in a cursor set, click  or  Button, it is possible to move the cursor position, rotate "ADJUST" Knob to adjust the Amplitude of the output waveform, as shown below:

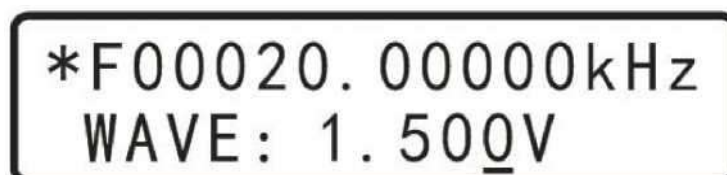


*F00020.00000kHz
WAVE: 05.00V

Figure 2-5

Wherein, 05.00V refers peak to peak. In this range setting function mode, the maximum amplitude of 20V, minimum 0.20V, the minimum step value 0.01 (10mV);

as shown in a **Figure 2-6**, press  -20dB attenuation state of the incoming signal, the output signal of a maximum of 2.000V, the minimum value of 0.005V, the minimum step is 0.001V (1mV).



*F00020.00000kHz
WAVE: 1.500V

Figure 2-6

4. Setting bias of CH1

In the main interface, press  or , adjusted to offset adjustment options shown in **Figure 2-7**, then click  + , the "*" are switched to the second row, click  or  to move the cursor, then "ADJUST" Knob to adjust the Offset parameter.

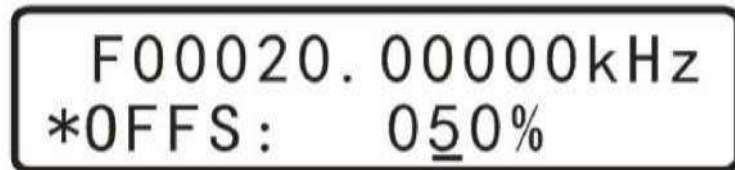


Figure 2-7

5. Duty cycle setting CH1

In the main interface, press  or , adjusted to duty cycle adjustment options shown in **Figure 2-8**, then click  + , the "*" are switched to the second row, click  or  to move the cursor, then "ADJUST" knob to adjust the Duty parameter.

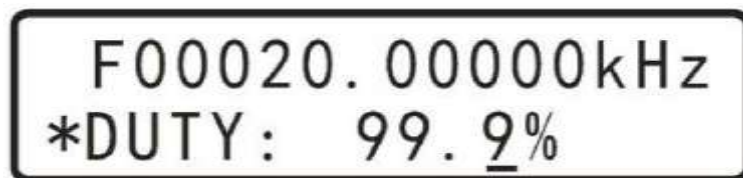


Figure 2-8

6. Setting the phase difference between the two channels

In the main interface, press  or , adjusting the phase adjustment options shown in **Figure 2-9**, then click  + , the "*" are switched to the second row, click  or  to move the cursor, then "ADJUST" Knob to adjust the Phase parameter, you need special note is the Phase difference only in the same frequency CH1 and CH2 frequency when it makes sense.

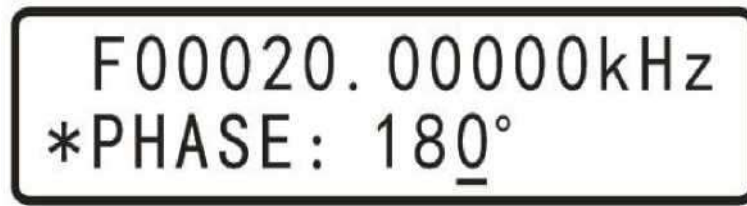


Figure 2-9

7. Setting the display unit of frequency

In the main interface, press  or , adjusted to the display Frequency Unit adjustment options shown in **Figure 2-10**, then click  + , the "*" are switched to the second row, then click  switching frequency units Hz, kHz, MHz.



Figure 2-10

8. Tracking

Tracking function is used to synchronize the frequency of CH2 to CH1, and users can set the amplitude of the tracking and duty track.

In the main interface, press  or , adjusted to Tracking options shown in **Figure 2-11**, then click  + , the "*" are switched to the second row, and then click  switched ON or OFF state. When the tracking function is turned frequency automatic tracking frequency channel CH2 CH1 channel. Also, before turning if tracing, CH1 and CH2 channel amplitude is the same tracking feature is turned on CH1 and CH2 frequency automatic tracking; if the duty is the same as CH1 and CH2 are turned on after tracing CH1 and CH2 account empty than automatic tracking.



Figure 2-11

9. External signal input port selection

Select the input signal Ext.IN for input AC signal, Select TTL.IN for selecting the input digital signal wave.

In the main interface, press  or , adjust to the Input port selection page shown in **Figure 2-12**, then click  + , the "*" are switched to the second row, then click  Key switch input port selection Ext.IN or TTL.IN.



Figure 2-12

10. Measurement function

In selecting the input source later, the input signal can be measured a variety of parameters.

(1) In the main interface, press  or , adjusted to the Measurement function selection page shown in **Figure 2-13**, then click  + , the "*" are switched to the second row, then click  Bond switch the output port selection measurement object: FREQ. (Frequency), COUNTR (Counting function), POS-PW (Postive Pulse Width), NEG-PW (Negative Pulse Width), PERIOD (Cycle), DUTY (Duty Cycle).

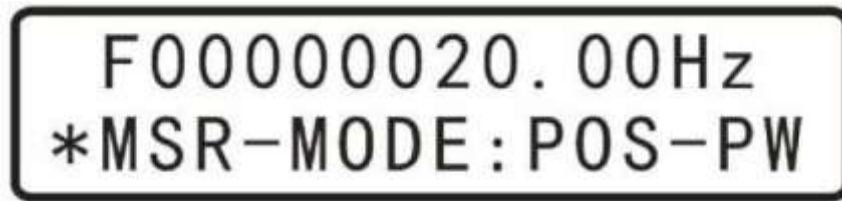


Figure 2-13

(2) After determining the measurement object, click , enter Gate time selection page as shown in **Figure 2-14**.

(3) Click  Bond, select a different gate time 10s, 1s, 0.1s, 0.01s, different gate time on the frequency measurement accuracy and measurement speed.



Figure 2-14

(4) Gate time is determined, click , enter the Measurement results display as shown in **Figure 2-15**. The page can display the input measurement results, such as frequency, duty cycle, pulse width, and so the cycle parameters.

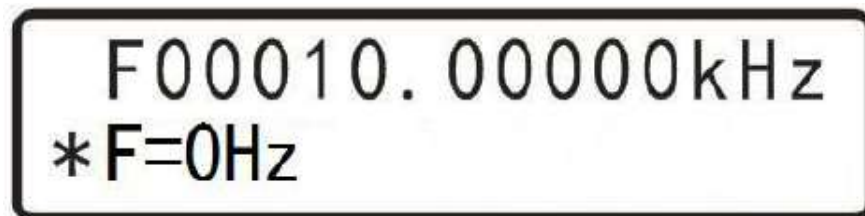


Figure 2-15

11. Sweep function

(1) In the main interface, press  or , adjusted to the Sweep function selection page shown in **Figure 2-16**, and then adjust the start frequency 5kHz.



Figure 2-16

(2) Then click  Into the cut-off frequency Sweep function settings page, and then adjust the cutoff frequency of 10kHz. As shown in **Figure 2-17**.

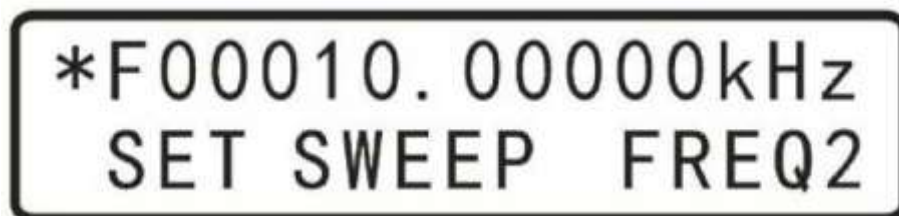


Figure 2-17

(3) Then click  enter Sweep time setting page, then click  +  the are switched to the second row, rotation "ADJUST" Knob to adjust the sweep time, frequency arbitrarily set the time range between 1-600s, shown in the lower set sweep time 10s:



Figure 2-18

(5) Then click  enter Sweep Mode selection screen shown in **Figure 2-19**, click  Bond select the frequency scan mode o there are two ways to scan frequency, LINE (linear scanning) and LOG (logarithmic sweep).

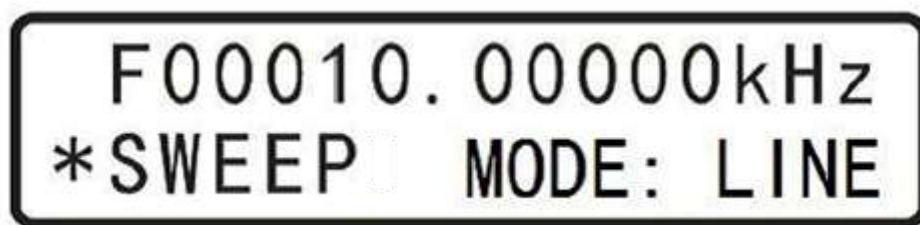


Figure 2-19

(6) After determining the scan mode, click  enter Sweep control page shown in **Figure 2-20**, then click  on or off sweep function.



Figure 2-20

12. Save / Load function

In the main interface, press  or  Enter the parameter to Save the page, and then click  +  press the key combination of "*" consumption is adjusted to the second row in **Figure 2-21**, then rotate the "ADJUST" Knob to choose the save location, and finally click the  Button will save the data in the set position. This machine a total of 10 sets of parameters stored addresses M0-M9. Machine is turned on by default to read M0 address parameters.

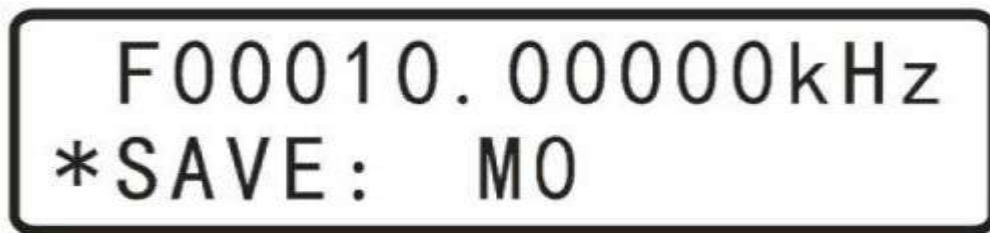


Figure 2-21

In the main interface, press  or  etnter the parameter to call up the page, then click  +  Press the key combination of "*" consumption is adjusted to the second row in **Figure 2-22**, then rotate "ADJUST" Knob to adjust the save location, and finally click the  Button the data from the set position. This machine a total of 10 sets of parameters stored addresses M0-M9. Machine is turned on by default to read M0 address parameters.

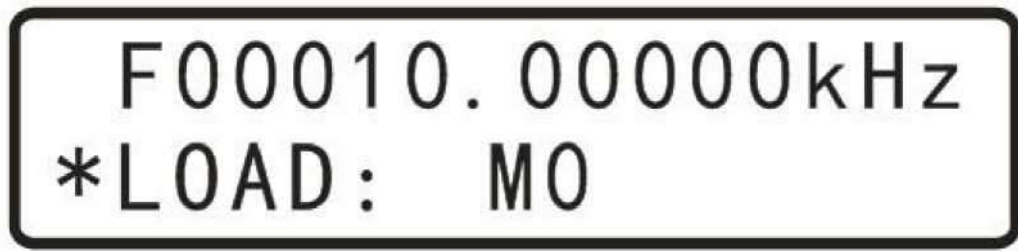


Figure 2-22

13. Reverse function

This function can quickly achieve the corresponding channel of the output waveform phase 180 degrees change.

In the main interface, press  or  Key, the adjustment to the measurement function selection page as shown in **Figure 2-23** shows, then click the  Button, open the Reverse function as shown in **Figure 2-24** shows.



Figure 2-23



Figure 2-24

14. Burst function

This function can be achieved CH2 channel burst CH1 channel output, to achieve the burst function is the premise of the CH1 channel set waveform frequency to be greater than the CH2 channel set waveform frequency, turn on the trigger function, the

CH2 channel waveform for each cycle start position will trigger CH1 channel output a pulse wave.

In the main interface, press the  or  Key, adjust to the Burst function control page as shown in **Figure 2-25** shows,



Figure 2-25

then click on the  Button, turn on the Burst function, as shown in **Figure 2-26**.



Figure 2-26

15. 4-channel TTL output function

The present machine can be output simultaneously 4-way TTL, in CH1 and CH2 in different steps of the state of TTL1 AND TTL3, AND TTL4 and the CH1 channel synchronization, the duty ratio of the CH1 decision; TTL2 and CH2 synchronization, the duty ratio of the CH2 decision. If CH1 and CH2 sync, in this case TTL1 AND TTL2 AND TTL3, AND TTL4 Quad TTL at the same time synchronization, the phase from the CH1 and CH2 the phase difference between the decision.

16. Calibration function

The machine in the factory when we had calibrated well, such as the need for their calibration can be to the manufacturers advice.

4. PC Instructions

1. Install Software

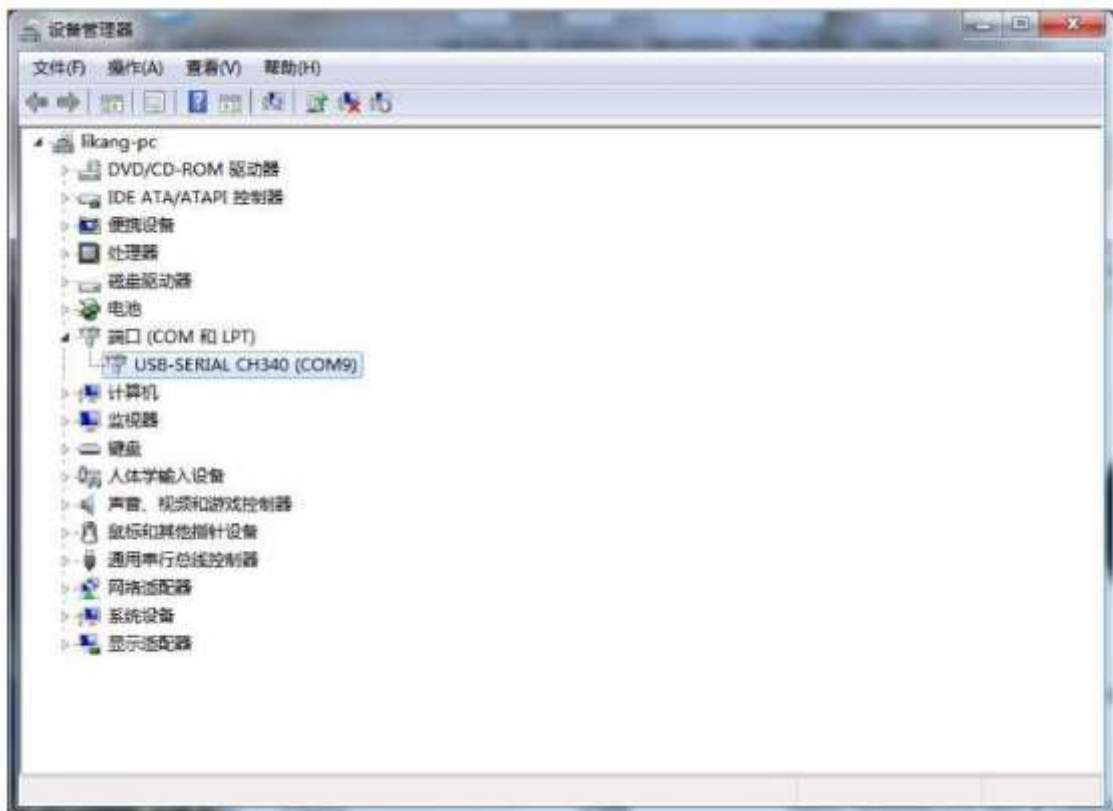
Step 1: Install the runtime software **visa540_runtime.exe**

Step 2: Install **CH341SER** the **SETUP.exe** serial to USB drivers

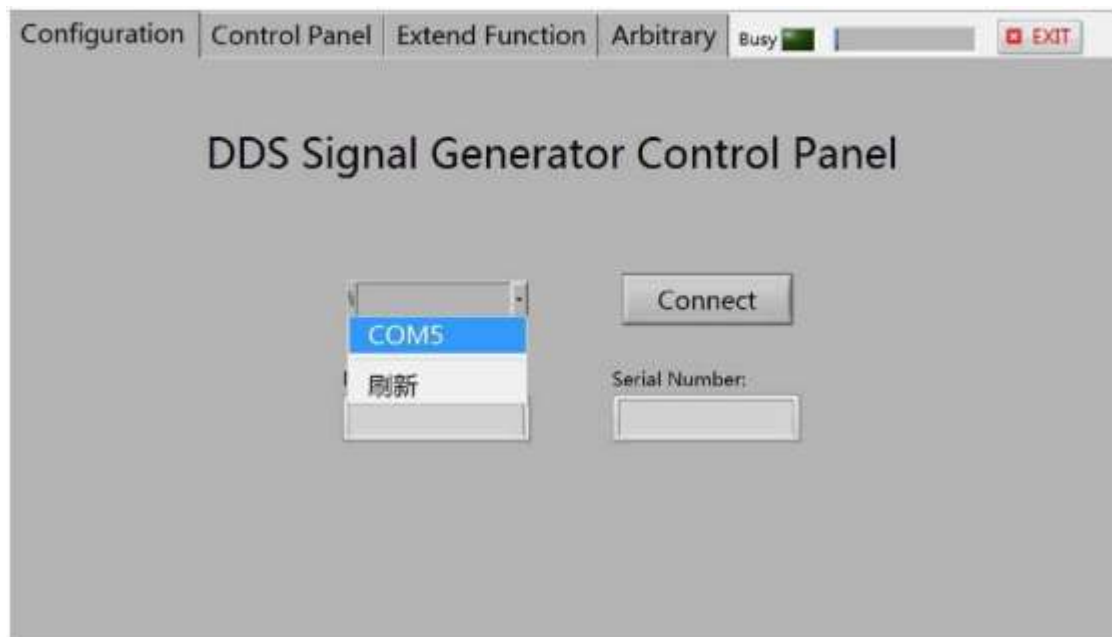
Step 3: Install signal generator program **MHS-5200A_en5.01.exe**

2. Online

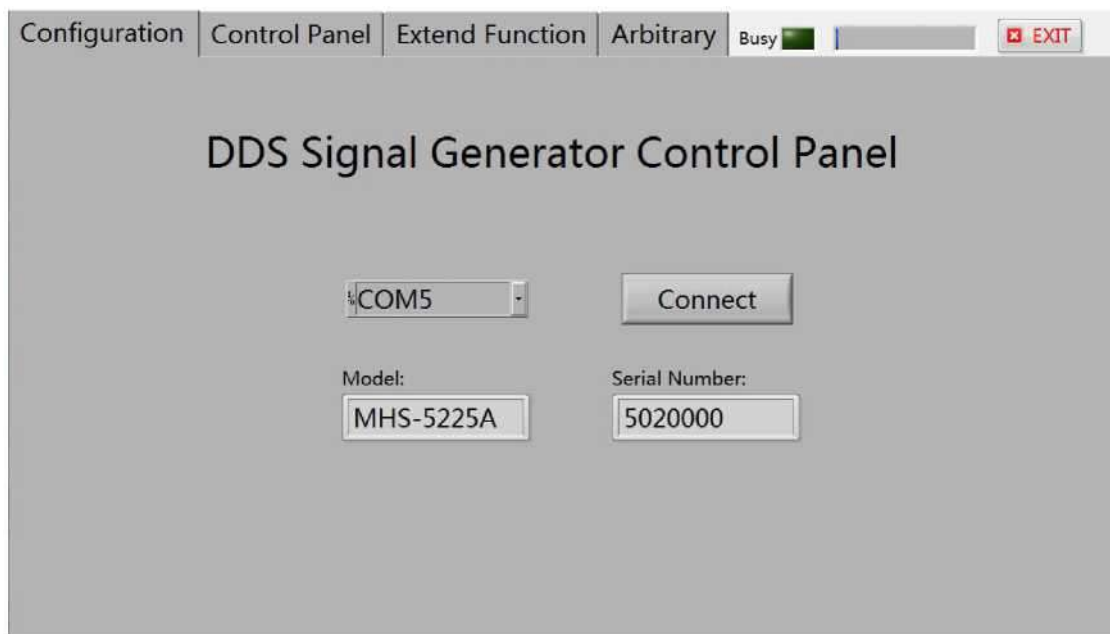
Step 1: right-click on the Computer > Property > Hardware > Device manager > Ports (COM & LPT)



Step 2: choose the corresponding point on line after a serial port

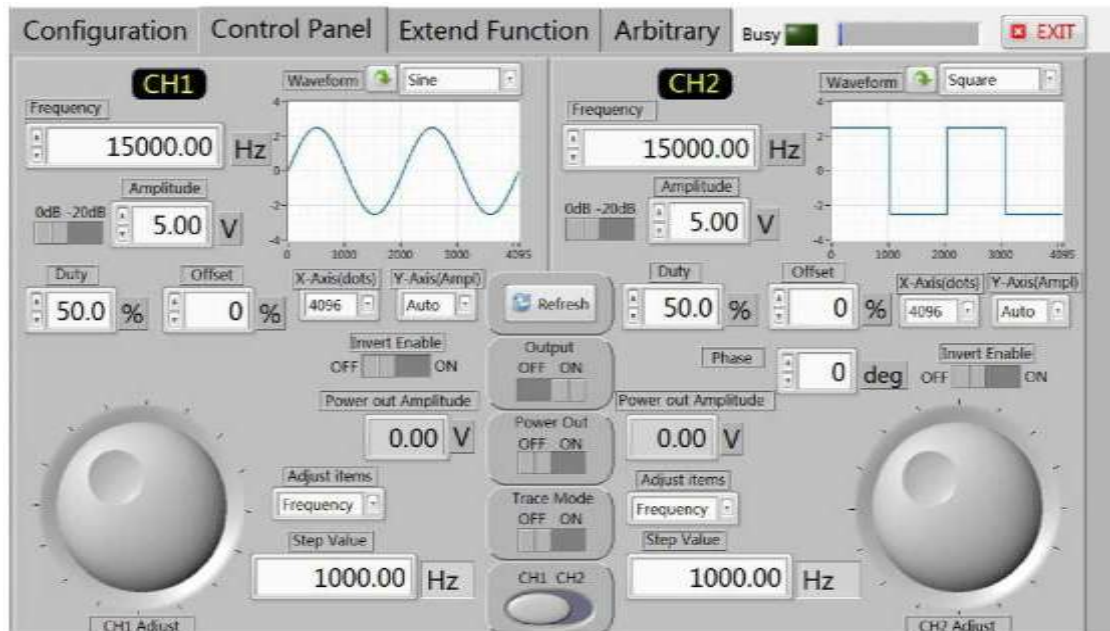


Step 3: online success

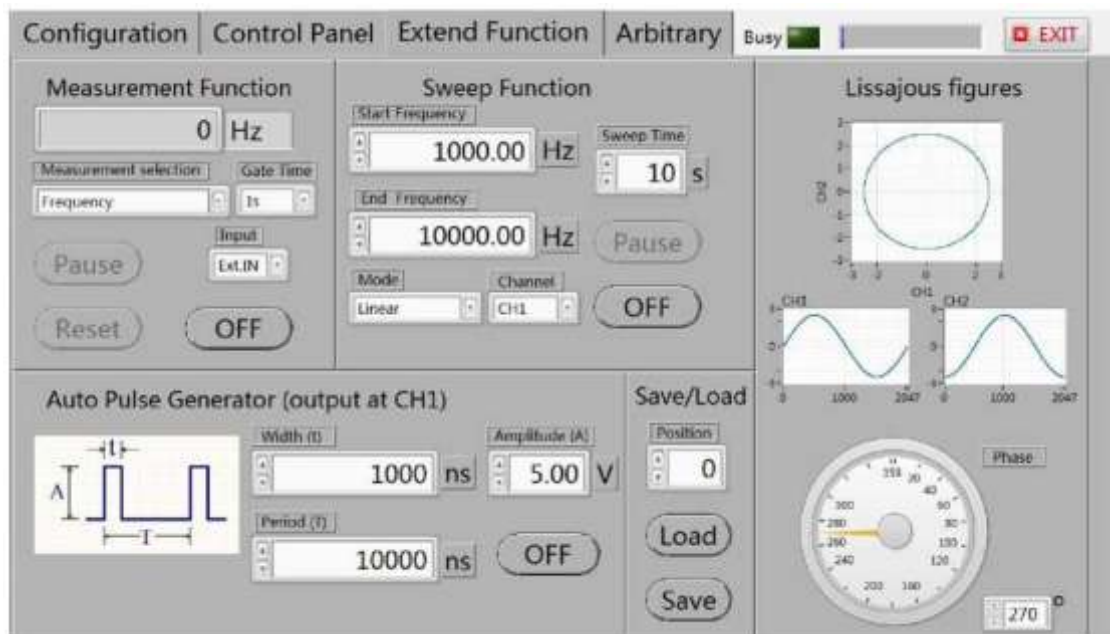


1: standard output waveform

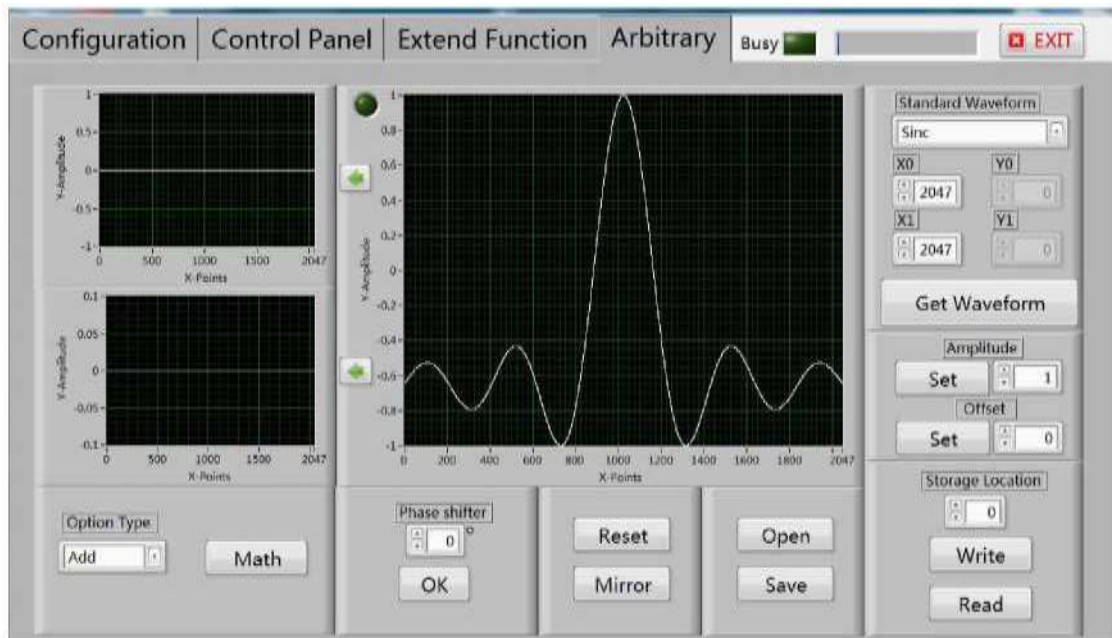
(1) the control signal output normal function, display boxes can real-time display current output waveform;



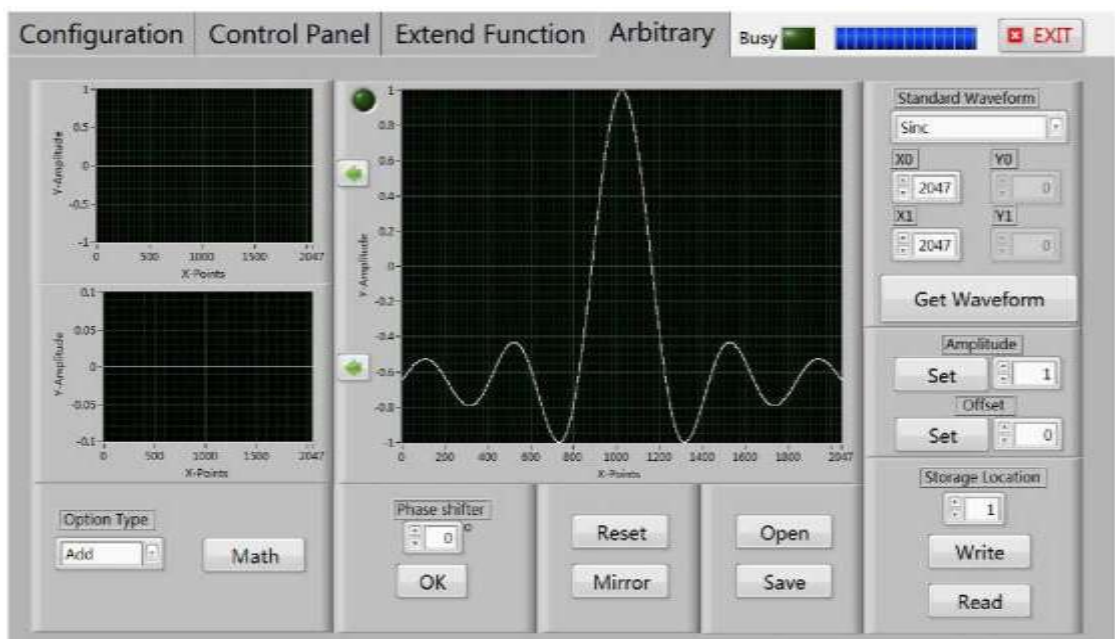
(2) measure the external signal function, can also be precise control of pulse width and cycle;



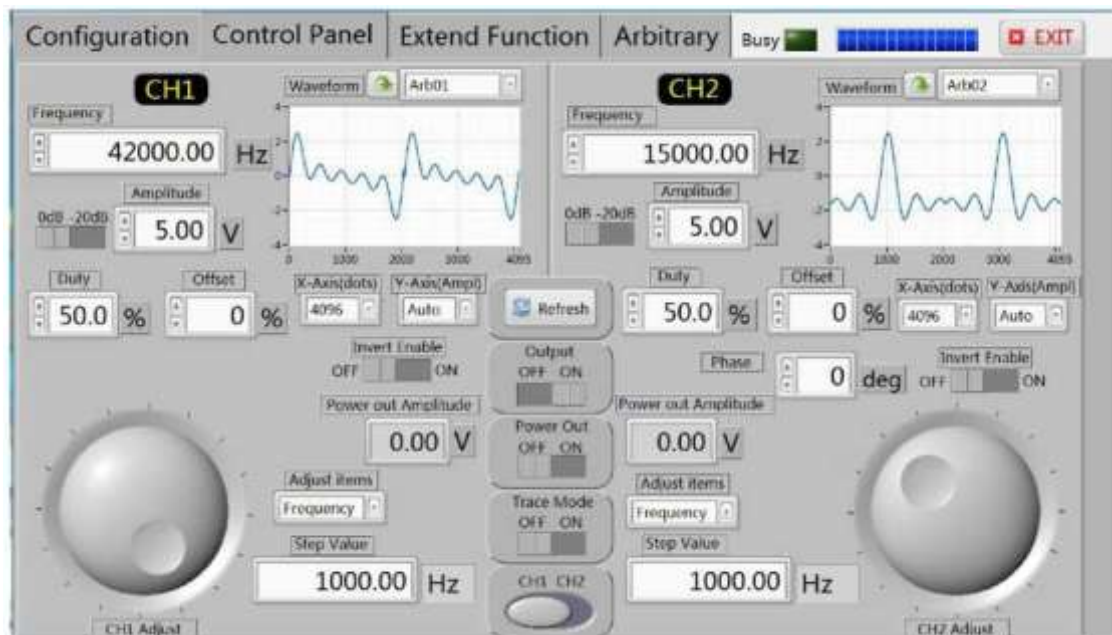
(3) connect the upper machine, switch to expand page, can control the output arbitrary waveform, we have some regular arbitrary wave, default of the currently selected is the medical pulse wave;



(4) keep the location choice, this series of machine with 0-15 waveform save location, select position 1 here, and then click on "write"; Blue Bar said written progress;



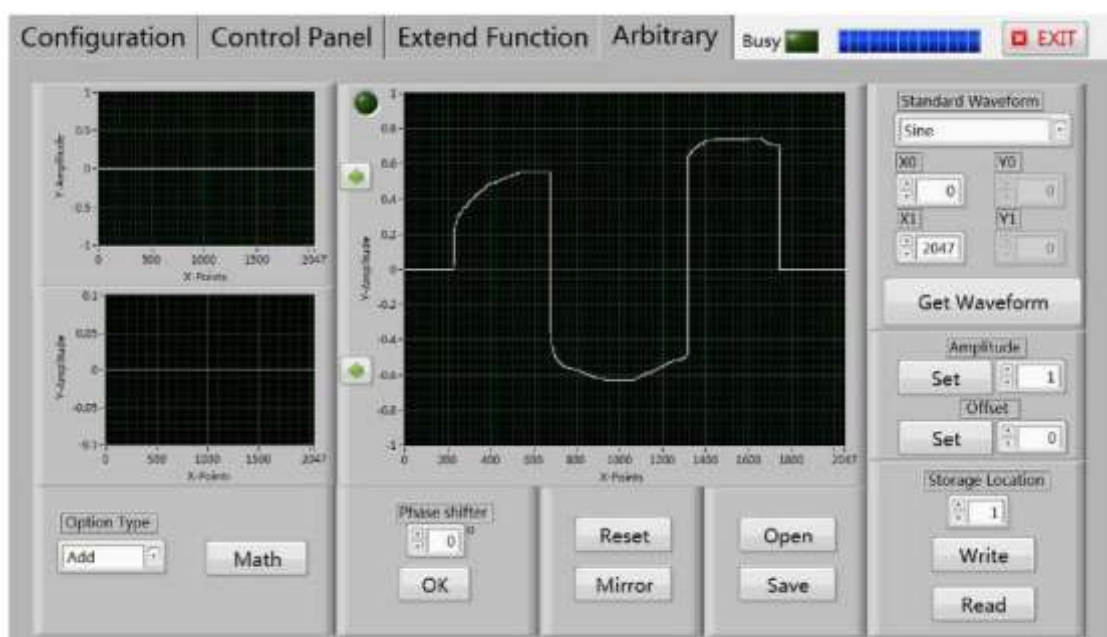
(5) after the completion of the write CH1 channel Arb1 v, the signal control page below:



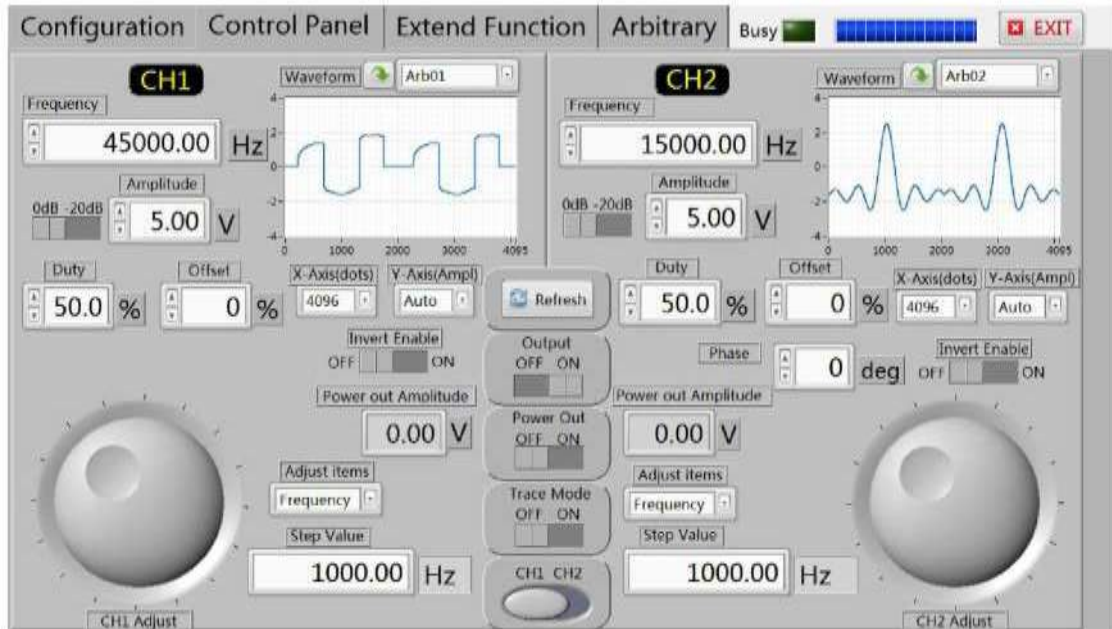
2: hand-painted arbitrary wave

(1) the upper machine and machine good connection, ensure the normal order of the communication between machine and computer.

(2) the software switch to expand the function pages, and then move the mouse to waveform display area, and then press the left mouse button to start drawing, release the left mouse button to stop drawing.

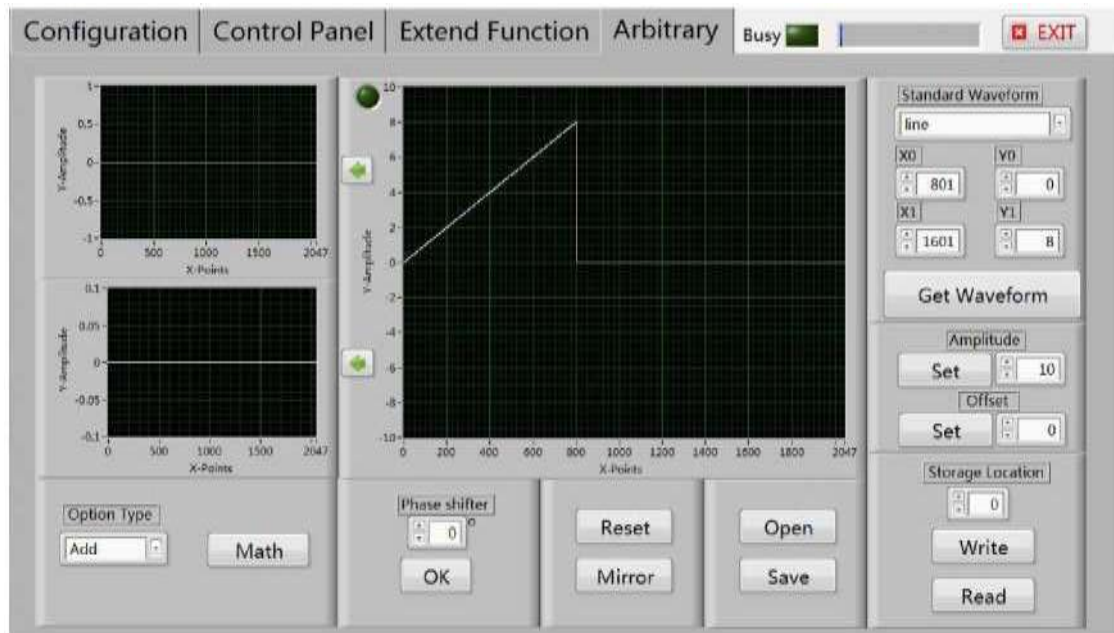


(3) to save the location of the waveform in the corresponding position, and then signal control page will bring up the map the location of the waveform in the preservation of the.

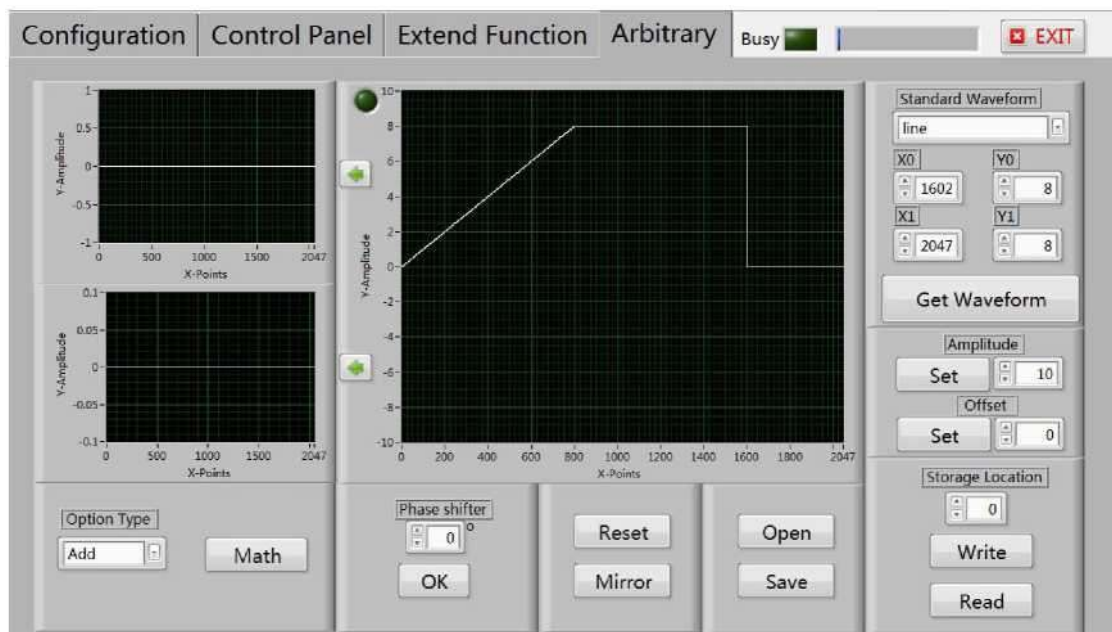


3: Hand draw a trapezoidal wave

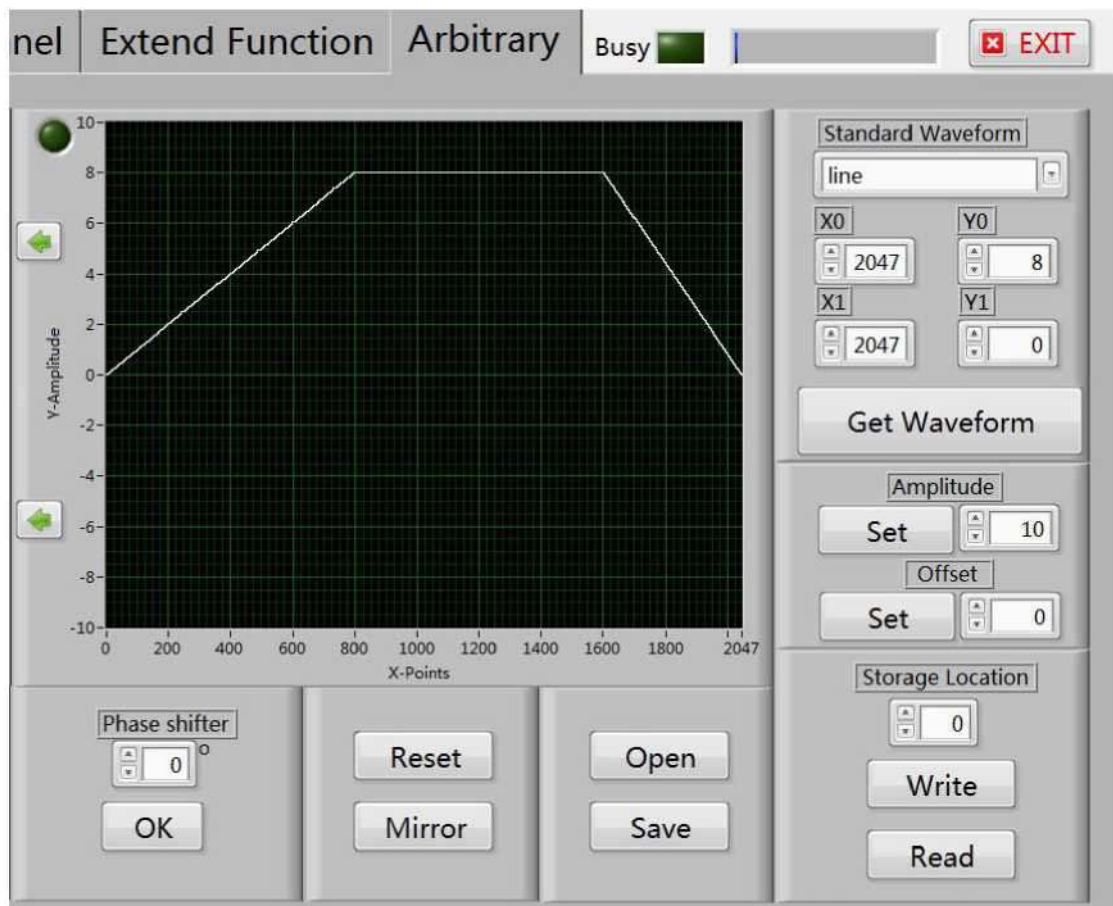
- (1) The upper machine and machine good connection, ensure the normal order of the communication between machine and computer.
- (2) Hand-painted trapezoidal in standard waveforms when choosing a straight line;
- (3) Set the starting point for 0, termination point 800; the starting range 0, terminate the magnitude 8
- (4) Click on the generated waveform generated waveform as follows:



(5) In setting up starting point, end points, 801, 1601; starting magnitude 8, termination of magnitude 8, then click the generated waveform, generated waveform is as follows:



(6) In setting up starting point, end points, 1602, 2047; starting magnitude 8, termination by 0, and then click the generated waveform, forming waveform as shown:



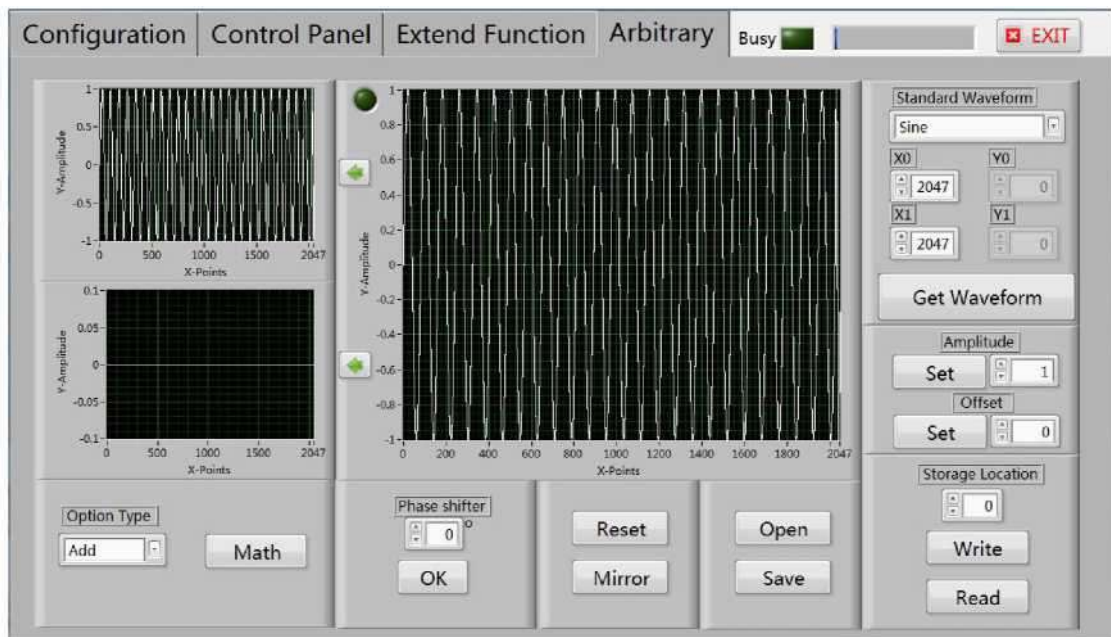
(7) Then put the waveform is saved in the corresponding position, and then signal control page will bring up the map the location of the waveform in the preservation of the.

This is the way how to draw a trapezoidal wave, one instance, in this way can draw other types of waveforms.

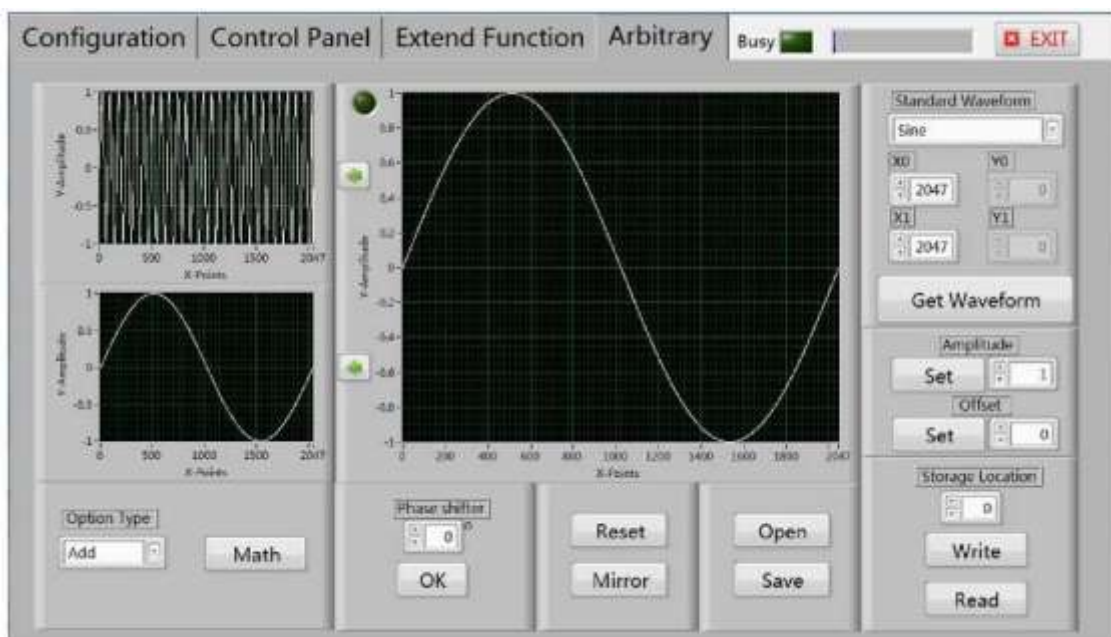
4. Draw a amplitude modulation waveform

(1) The upper machine and machine good connection, ensure the normal order of the communication between machine and computer.

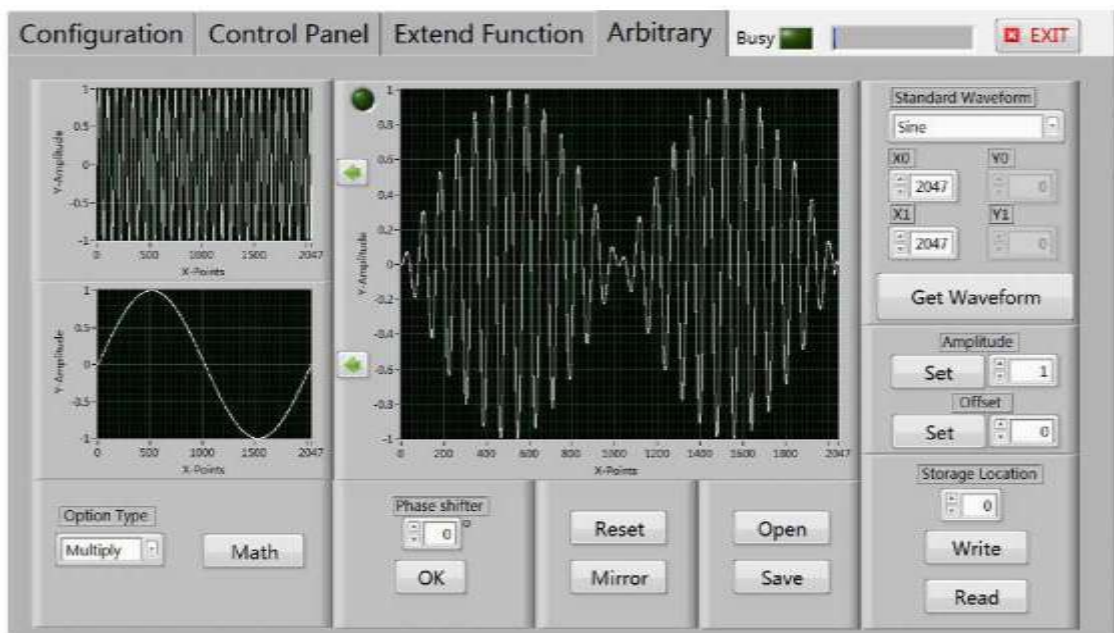
(2) In the standard waveform selection box choose sine wave, the starting point for writing 0; end point written 80, and then click the generated waveform button in a row, and then click the green arrow to the right form waveform as shown:



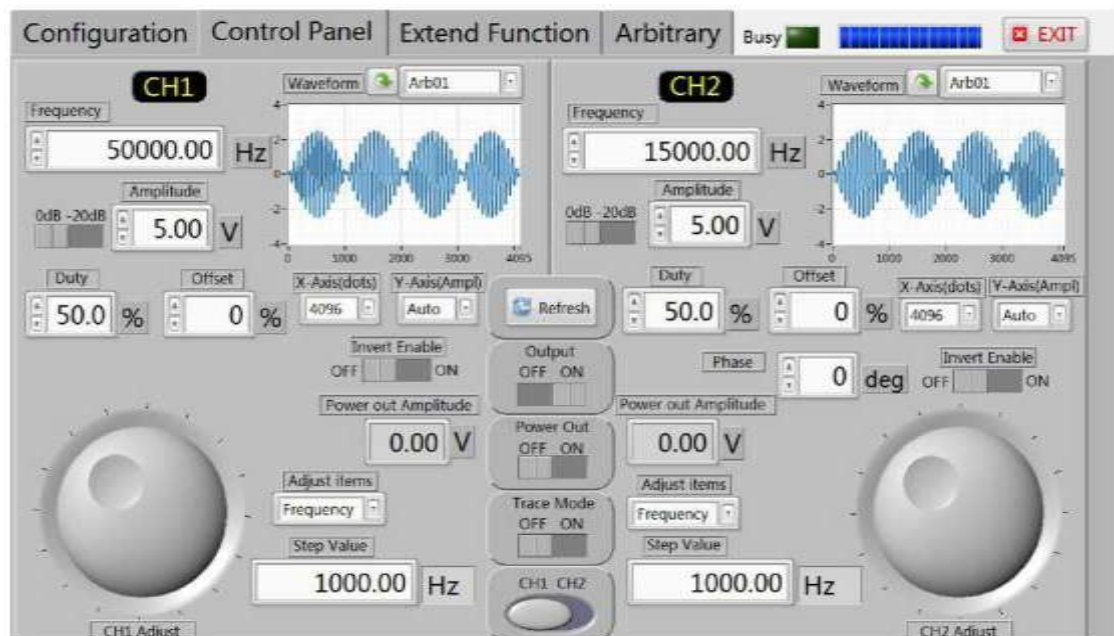
(3) Set the starting point 0; end point 2048, and then click "generate the waveform" button. Click the green arrow to the right generated waveform as shown:



(4) Select operation type product, and then click operation button, generates the amplitude modulation waveform.



(5) Then to save the location of the waveform on 01 location, and then will draw waveform signal control page in the preservation of 01 out of position.



5. Care and maintenance

1. Make sure the input power adapter correctly, the machine uses DC5V power adapter.
 2. The instrument display on the LCD module is fragile, perishable items, please do not slam and near chemicals to prevent corrosion. When you feel the liquid surface dust and dirt, wipe with a soft cloth carefully.
 3. The working temperature of (-10 ~ +50) °C, Storage temperature (-20 ~ +70) °C, and the instrument in a dry environment.
 4. Do not attempt to disassemble the equipment, destroy the package will void the warranty. The instrument is there are no user-serviceable parts, repairs may only repair outlets or by specifying the return factory.
 5. Avoid lighted candles, a water cup, corrosive chemicals and other unsafe items placed on the surface of the instrument, so as not to cause damage to the instrument.
 6. The display screen are easy to pollution, fragile device, do not touch the hand as well as external and collision, avoid children play this instrument.
 7. Do not move the instrument to avoid severe irreparable damage to the internal circuit when the instrument is working properly.
- Exclude the above problem re-power the instrument still does not work, please contact your supplier!

Warranty and after-sales service

Thank you to buy Ming WO electronic products. To maximize understanding and using your new product functionality, we recommend that you take the following steps:

1. Read the safety and effective use of the guide.
2. Read the warranty terms and conditions.

Warranty conditions:

Instrument shipment date of the warranty period is one year. During the warranty period the company according to the situation choose to fault the instrument for repair or replacement. If you need maintenance, please contact after sales and will present the product by mail to my company.

The following case is not in the warranty scope:

The user operation or improper maintenance; using the user's own supplied software or power supply interface; without the permission of the instrument for disassembly for repair.